

Knowing When the Model Doesn't Know

Conformal Uncertainty Quantification for Mammographic Malignancy Classification

ApeX | ACM 40960 Project 8: Disease Modelling | Supervisor: Dr. Sarp Akçay | University College Dublin

A DenseNet-121 mammography classifier is wrapped with split conformal prediction, turning uncalibrated scores into label sets that are guaranteed to contain the true diagnosis 95% of the time — automatically flagging ambiguous cases for radiologist review.

1. The Problem

A standard classifier reports one label and an **uncalibrated** confidence. In breast-cancer screening that is not enough: the clinician needs to know *when to distrust the model*. We need predictions with a **stated, guaranteed error rate** — not a number that merely looks like a probability.

2. Approach

Split conformal prediction converts any classifier's scores into a **label set** with a distribution-free coverage guarantee:

$$\Pr(Y_{\text{true}} \in \mathcal{C}(X)) \geq 1 - \alpha$$

No assumption is made on model calibration — only that calibration and test data are **exchangeable**.

Pipeline

- DenseNet-121, fine-tuned on CBIS-DDSM (benign vs. malignant)
- Patient-level splits — no leakage into calibration
- Threshold $\hat{q}$ set on a held-out calibration set
- Three scores compared: **LAC**, **APS**, **RAPS**

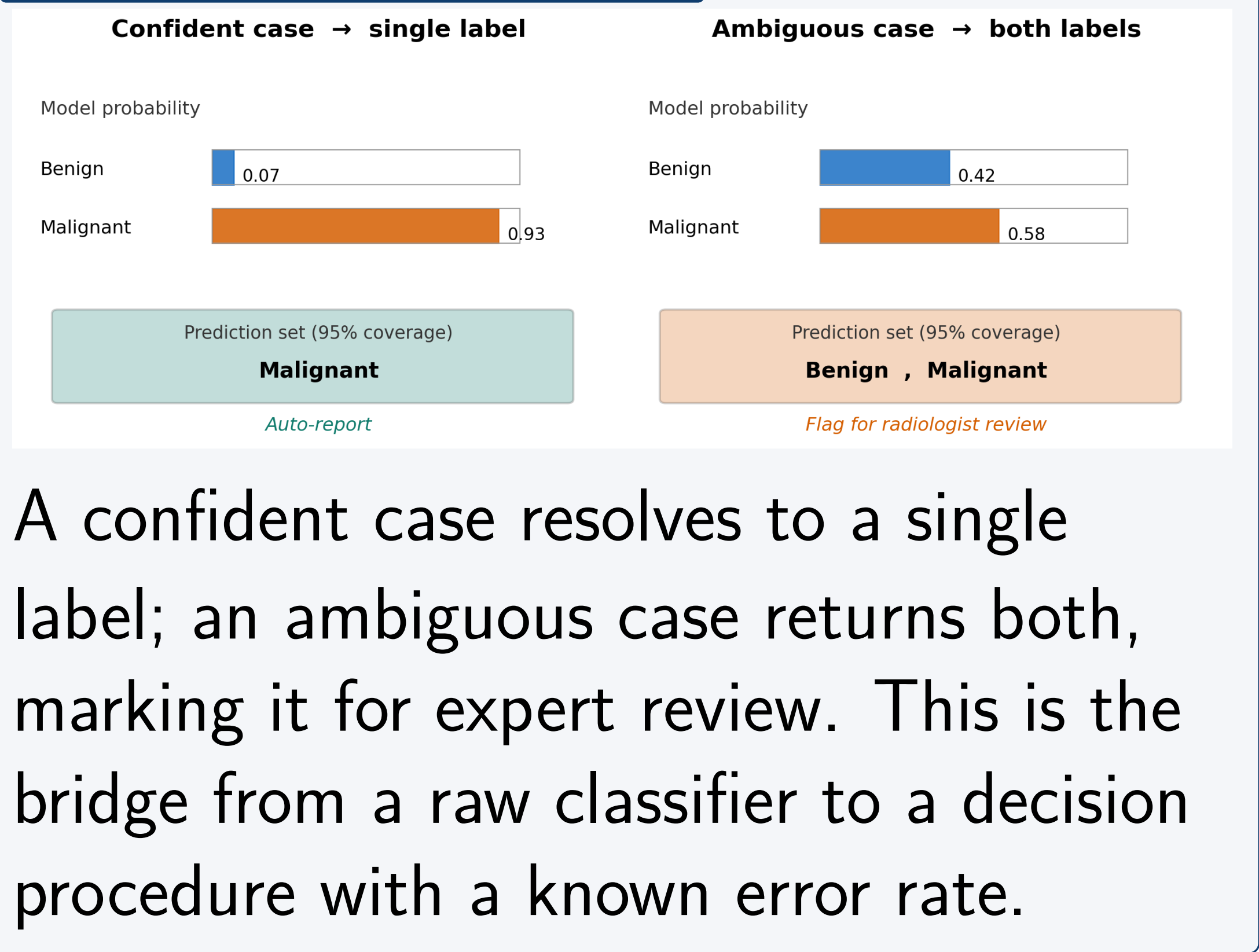
3. How the Set is Built

Every class whose nonconformity score falls below $\hat{q}$ enters the set.

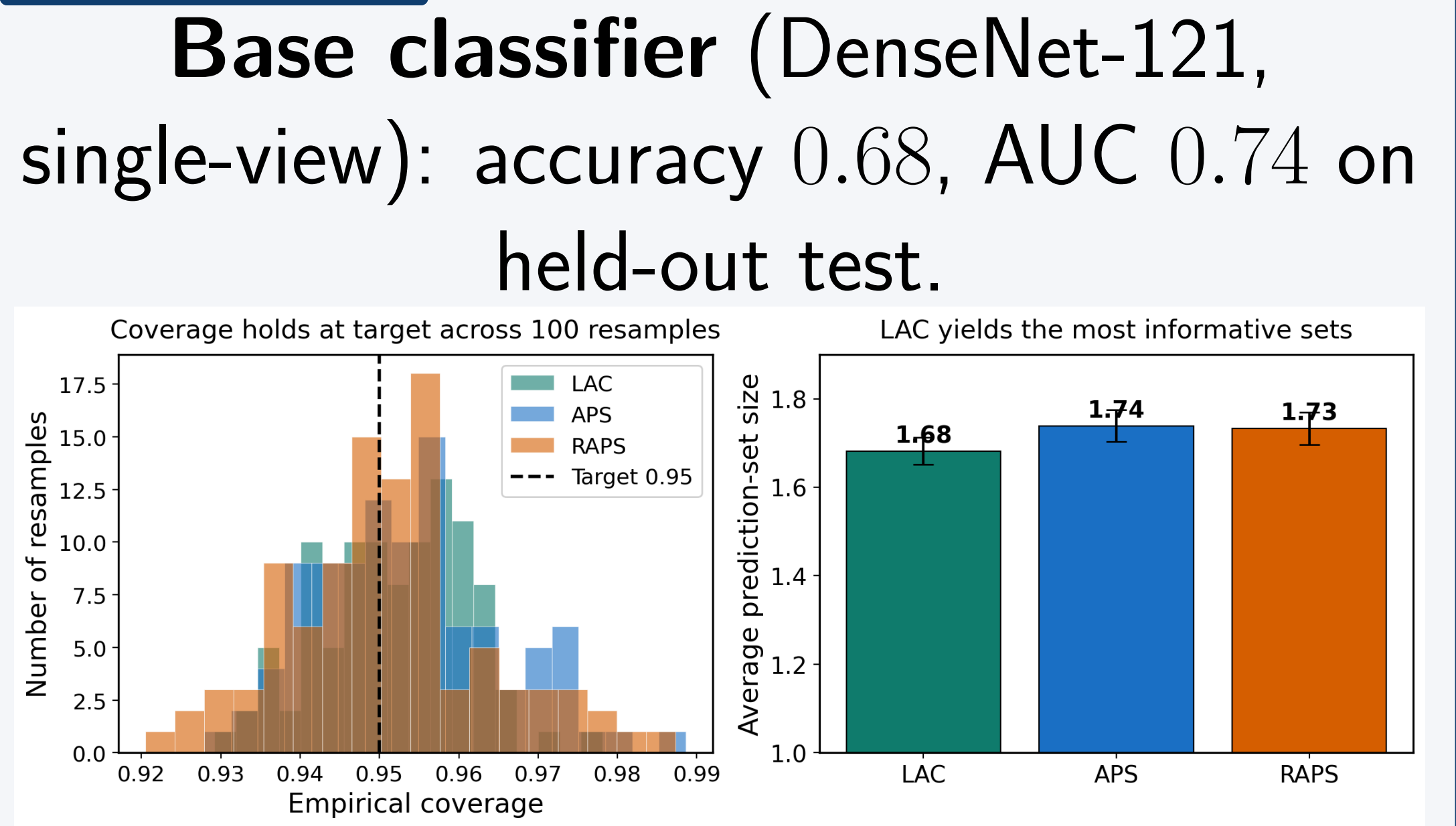
- **One label** → model is confident → auto-report
- **Both labels** → model is unsure → escalate

The set size *is* the uncertainty signal.

4. The Clinical Idea



5. Results



Left: across 100 calibration/test resamples all three methods sit on the 0.95 target line — the guarantee holds empirically. **Right:** LAC produces the smallest sets (1.68), resolving 32% of cases to a single confident label while flagging the rest for review.

Method	Coverage	Set size	Singleton
LAC	0.953	1.68	0.32
APS	0.954	1.74	0.26
RAPS	0.952	1.73	0.27

Mean over 100 resamples, $\alpha = 0.05$.

6. Why It Matters

- **Guaranteed coverage** without trusting the model's raw probabilities — critical where errors are costly.
- **Automatic triage:** single-label cases can be auto-reported; two-label cases route to a radiologist.
- **Model-agnostic:** the conformal layer sits on *any* classifier unchanged.

7. Method is Verified

The coverage guarantee is validated **independently of any real model** on synthetic data across 200 trials, confirming all three scores hold $\geq 1 - \alpha$ coverage before any mammography data is used. This separates a correct *implementation* from a fortunate *result*.

8. Scope & Future Work

- **Single-view** here; the conformal layer transfers directly to a multi-view bilateral-fusion backbone.
- **Binary** task; code already handles $K > 2$ for BI-RADS grading.
- **Cross-dataset** shift (e.g. VinDr-Mammo) needs weighted / Mondrian conformal variants — exchangeability would otherwise break.

Code & References Repository

Full pipeline, verification harness, and reproducible Kaggle notebook.

Angelopoulos & Bates (2021) · Romano et al. (2020) · Angelopoulos et al. (2021, RAPS) · Sawyer-Lee et al.



Scan for code